TONGSEOK LIM

◇ https://tlim0213.github.io ◇ lim336 at purdue dot edu ◇ November 9, 2026

ACADEMIC CAREER

- **Purdue University, West Lafayette, IN, USA** — Aug 2020 – present
  Assistant Professor in Quantitative Methods, Mitch Daniels School of Business
  Homepage: https://tlim0213.github.io
- **Fields Institute, Toronto, Canada** — Mar – Jul 2019 & Mar – Jul 2020
  Fields Research Fellow
- **ShanghaiTech University, Shanghai, China** — Sep 2018 – Jun 2020
  Assistant Professor of Mathematics
- **University of Oxford, Oxford, UK** — Oct 2016 – Aug 2018
  Postdoctoral Research Assistant
- **Technische Universität Wien, Vienna, Austria** — May – Sep 2016
  Postdoctoral Research Assistant
- **University of British Columbia, Vancouver, Canada** — Sep 2011 – May 2016
  PhD in Mathematics
- **Citizenship: Republic of Korea | Permanent Residency: United States**

RESEARCH INTERESTS

Optimization and mathematical foundations of data science; optimal transport, Schrödinger bridges, and Gibbs variational methods; data integration, statistical learning, and decision-making under uncertainty.

PUBLICATIONS

- **On the structure of optimal martingale transport in higher dimensions**. *PhD thesis*. Link
- with N. Ghoussoub and Y-H. Kim. **Structure of optimal martingale transport in general dimensions**. *Annals of Probability*, 2019. Journal
- with M. Beiglböck and J. Obłój. **Dual attainment for the martingale transport problem**. *Bernoulli*, 2019. Journal
- **Optimal martingale transport between radially symmetric marginals in general dimensions**. *Stochastic Processes and their Applications*, 2020. Journal
- with N. Ghoussoub and Y-H. Kim. **Optimal Brownian stopping when the source and target are radially symmetric distributions**. *SIAM Journal on Control and Optimization*, 2020. Journal
- with S. Eckstein, G. Guo and J. Obłój. **Robust pricing and hedging of options on multiple assets and its numerics**. *SIAM Journal on Financial Mathematics*, 2021. Journal arXiv
- with K. Kang, H. Kim and G. Seo. **Uniqueness and characterization of local minimizers for the interaction energy with mildly repulsive potentials**. *Calculus of Variations and Partial Differential Equations*, 2021. Journal arXiv
- with R.J. McCann. **Isodiametry, variance, and regular simplices from particle interactions**. *Archive for Rational Mechanics and Analysis*, 2021. Journal arXiv
- with R.J. McCann. **On Fejes Tóth's conjectured maximizer for the sum of angles between lines**. *Applied Mathematics and Optimization*, 2021. Journal arXiv

- with R.J. McCann. **Geometrical bounds for the variance and recentered moments**. *Mathematics of Operations Research*, 2022. Journal arXiv
- with R.J. McCann. **Maximizing expected powers of the angle between pairs of points in projective space**. *Probability Theory and Related Fields*, 2022. Journal arXiv
- with R.J. McCann. **On the cardinality of sets in $\mathbf{R^d}$ obeying a slightly obtuse angle bound**. *SIAM Journal on Discrete Mathematics*, 2022. Journal arXiv
- with C. Davies and R.J. McCann. **Classifying minimum energy states for interacting particles: Spherical shells**. *SIAM Journal on Applied Mathematics*, 2022. Journal arXiv
- with C. Davies and R.J. McCann. **Classifying minimum energy states for interacting particles: Regular simplices**. *Communications in Mathematical Physics*, 2023. Journal GitHub arXiv
- **Maximal monotonicity and cyclic involutivity of multiconjugate convex functions**. *SIAM Journal on Optimization*, 2023. Journal GitHub arXiv
- **Geometry of vectorial martingale optimal transportations and duality**. *Mathematical Programming* (series A), 2024. Journal GitHub arXiv
- with E. Koo. **Node classification in networks via simplicial interactions**. *IEEE Transactions on Neural Networks and Learning Systems*, 2025. Journal GitHub arXiv
- with E. Koo. **Node classification via simplicial interaction with augmented maximal clique selection**. *Neurocomputing*, 2025. Journal GitHub arXiv
- with J. Hiew, B. Pass and M. Souza. **Data denoising with self consistency, variance maximization and the Kantorovich dominance**. *SIAM Journal on Mathematics of Data Science*, 2026. Journal arXiv
- with E. Koo and J. Sohn. **Statistical matching via Schrödinger bridge beyond conditional independence**. *The 40th Conference on Neural Information Processing Systems*, to appear. GitHub arXiv

PAPERS UNDER REVIEW

- with K. Nam and J. Sohn. **Monotone curve estimation via convex duality**. GitHub arXiv
- with C. Che and Y. Sun. **Dual attainment in multi-period multi-asset martingale optimal transport and its computation**. arXiv
- **Stable allocations, Unstable payment paths: The Shapley value and voluntary coalition formation**. GitHub arXiv
- with W. Ha and K. Nam. **Statistical guarantees for multimodal contrastive learning via pointwise mutual information estimation**. GitHub
- with G. Seo. **Uniqueness and shape of simplex ground states for aggregation–diffusion energies with mild repulsion**. GitHub arXiv
- with Y. Zhang. **Random basket quotes: minimax distributional recovery and price estimation**. GitHub

WORK IN PROGRESS

- with W. Ha and K. Nam. **Learning implicit multivariate interactions via Gibbs relaxation**.
- with J. Hiew, B. Pass and M. Souza. **Dimension reduction in martingale optimal transport: geometry and robust option pricing**. GitHub arXiv
- with C. Che, P. Das and Y. Sun. **Observable-augmented martingale optimal transport for robust pricing and hedging of multi-asset exotics**.
- **Optimal exercise decision of American options under model uncertainty**. GitHub arXiv

· **Axiomatic and probabilistic foundations for the Hodge-theoretic Shapley value**. GitHub
· with D. Jeong and E. Shin. **Robust intervention in networks**. GitHub arXiv
· with D. Jeong and E. Shin. **Value allocation in coalition networks**. preliminary version on arXiv
· with J. Kil. **From pairs to teams: sparse flexibility with complementary resources**.

INVITED TALKS & LECTURES

· Aug 2026. *9th International Conference on Econometrics and Statistics.* Ryukoku University, Japan. Link
· Jul 2026. *2026 Asian Quantitative Finance Conference.* Lahan Hotel Pohang, Korea. Link
· Aug 2025. *Workshop on Quantum Analysis & Geometry.* Lakai Sandpine Resort, Korea. Link
· Jul 2025. *Statistics seminar.* The Chinese University of Hong Kong.
· May 2025. *Workshop on Mathematical Finance at Seoul.* Seoul National University, Korea. Link
· Jul 2024. *Applied Mathematics seminar.* Nanyang Technological University, Singapore.
· Jun 2024. *Optimal Transport, Stochastic Analysis and Machine Learning.* KAIST, Korea. Link
· May 2024. *Workshop on Mathematical Finance at Jeju.* Jeju Island, Korea. Link
· Mar 2024. *Financial and Actuarial Mathematics seminar.* University of Michigan, USA. Link
· May 2023. *Conference on Approximation Theory and Beyond.* Vanderbilt University, USA. Link
· Apr 2023. *Midwest International Trade & Theory Conference.* University of Tennessee, USA. Link
· Mar 2023. *Finance seminar.* College of Business, Stony Brook University, USA.
· Mar 2023. *Eighth annual conference on network science and economics.* Virginia Tech, USA. Link
· Feb 2023. *Quantitative Methods seminar.* School of Business, Purdue University, USA.
· Feb 2023. *Analysis & PDE seminar.* Department of Mathematics, Purdue University, USA.
· Jan 2023. *2023 Joint Mathematics Meetings (JMM 2023).* Boston, USA. Link
· Jul 2022. *AI seminar.* Center for AI and Natural Sciences, KIAS, Korea. Link
· Jun 2022. *Colloquium seminar.* Kyungpook National University, Korea.
· May 2022. *Colloquium seminars I & II.* Pusan National University, Korea.
· May 2022. *AI seminar.* Center for AI and Natural Sciences, KIAS, Korea. Link
· Mar 2022. *Stochastic Mass Transports.* Banff International Research Station. Link Video
· Dec 2021. *Analysis & PDE seminar I & II.* Yonsei University, Korea.
· Oct 2021. *Game theory seminar.* College of business, KAIST, Korea.
· Sep 2021. *Mathematics & Economics joint seminar.* Washington University in St. Louis, USA. Link
· Dec 2020. *Canadian Mathematical Society Winter Meeting.* online. Link Video
· Feb 2020. *Colloquium.* Krannert School of Management, Purdue University, USA.
· Oct 2019. *Korean Mathematical Society (KMS) Annual Meeting.* Hong-Ik University, Seoul, Korea.
· Oct 2019. *Analysis & PDE Seminar.* Georgia Institute of Technology, USA. Link
· Apr 2019. *Analysis & Applied Math Seminar.* University of Toronto, Canada.
· Mar 2019. *Colloquium Seminar.* Pusan National University, Korea.
· Aug 2018. *Mathematics Seminar.* Seoul National University, Korea.
· Jun 2018. *Mathematical Finance Internal Seminar.* University of Oxford, UK.
· May 2018. *CMO Workshop 18w5080: Stochastic Analysis and its Applications.* Mexico. Link Video

· May 2018. *Mathematics Seminar.* University of Bath, UK.
· Apr 2018. *Mathematics Seminar.* University of Central Florida, USA.
· Dec 2017. *Vienna Seminar in Mathematical Finance and Probability.* University of Vienna, Austria.
· Dec 2017. *Mathematics Seminar.* Dublin City University, Ireland.
· Sep 2017. *Martingale Optimal Transport (and Friends).* University of Oxford, UK. Link
· Jul 2017. *Stochastic Processes and their Applications (SPA) 2017.* Moscow, Russia. Link
· May 2017. *Oxford-Princeton Workshop on Financial Mathematics & Stochastic Analysis.* UK. Link
· Jan 2017. *Mathematics Seminar.* Strasbourg University, France.
· Jan 2017. *Advances in Financial Mathematics.* Paris, France. Link
· Jan 2017. *Mathematics Seminar.* Ecole Polytechnique, Palaiseau, France.
· Jan 2017. *Paris Bachelier Seminar.* Paris, France. Link
· Nov 2016. *Mathematical Finance Internal Seminar.* University of Oxford, UK. Link
· Nov 2016. *Model Uncertainty & Robust Finance.* University of Milan, Italy.
· Jul 2016. *Mathematics Seminar.* Scuola Normale Superiore, Pisa, Italy.
· Jun 2016. *Mathematics Seminar.* Yonsei University, Seoul, Korea.
· Apr 2016. *Optimal Transportation, Equilibrium, and Applications to Economics.* NYU, USA. Link
· Mar 2016. *Skorokhod embeddings, Martingale Optimal Transport and their applications.* UK. Link
· Feb 2016. *Vienna Seminar in Mathematical Finance and Probability.* TU Wien, Austria. Link
· Sep 2015. *Mathematics Seminar.* Ecole Polytechnique, Palaiseau, France.
· Sep 2015. *European Summer School In Financial Mathematics.* Le Mans, France. Link
· Aug 2015. *KIAS CMC minischool on Analysis, Geometry, and Optimal Transport.* KIAS, Korea. Link

RESEARCH STUDENTS AND ALUMNI

PhD supervision

· Marcelo Cruz de Souza (Defended in July 2024)
· Joshua Zoen-Git Hiew (Defended in 2025 at Univ. of Alberta. supervised jointly with Brendan Pass)
· Junhyeok Kil (Started in January 2026)

MSc supervision

· Cameron Davies (currently PhD student at Univ. of Toronto, supervised jointly with Robert McCann)

GRANT PROPOSALS

NSF CAREER: "Gibbs Variational Methods for Data Fusion, Imputation, and Interaction Learning." Principal Investigator, $599,465. Submitted July 2026; under review.

TEACHING EXPERIENCE

Daniels School of Business, Purdue University 2020 – present
Assistant Professor in Quantitative Methods *West Lafayette, IN, USA*

· MGMT 30500–001,002,003 : Business Statistics Spring 2027
· MGMT 30500–002,003 : Business Statistics Spring 2026
· MGMT 69000 : Optimal Transport and Unsupervised Learning Spring 2025
· MGMT 47400–001,002 : Predictive Analytics Fall 2024
· MGMT 47400–001 : Predictive Analytics Spring 2024

· MGMT 47400–001,002 : Predictive Analytics — Fall 2023
· MGMT 47400–001 : Predictive Analytics — Spring 2023
· MGMT 47400–001,002 : Predictive Analytics — Fall 2022
· MGMT 69000 : Advanced Problems in Management – Optimal Transport Problems — Spring 2022
· MGMT 47400–001 : Predictive Analytics — Spring 2022
· MGMT 47400–001,002 : Predictive Analytics — Fall 2021
· MGMT 47400–001,002 : Predictive Analytics — Spring 2021

ShanghaiTech University — 2018 – 2020
Assistant Professor of Mathematics — *Shanghai, China*

· Stochastic Processes (for 3rd & 4th grades) — 2019
· Linear Algebra (for 2nd grades) — 2018

University of Oxford — 2018
Class Tutor — *Oxford, UK*

· Class Tutor for B8.3 – Mathematical Models of Financial Derivatives — 2018

University of British Columbia — 2011 – 2016
Teaching Assistant and Instructor — *Vancouver, BC, Canada*

· Instructor for MATH 101 – Integral Calculus with Applications to Physics and Engineering — 2016
· Instructor for Workshops – Direct student groups towards solving math problems — 2015
· TA for MATH 121 – Honours Integral Calculus — 2015
· TA for MATH 441 – Mathematical Modeling: Discrete Optimization Problems — 2014
· TA for MATH 424 – Classical Differential Geometry — 2012 – 2013

HONOURS & AWARDS

· Fields Research Fellowship. 2019 & 2020. University of Toronto, Canada.
· Four Year Doctoral Fellowship (4YF) recipient. 2011 – 2015. University of British Columbia, Canada.
· Kwanjeong Scholarship recipient. 2011 – 2016. Kwanjeong Educational Foundation, Seoul, Korea. Link
· Seoul Scholarship recipient. 2008 – 2010. Seoul Metropolitan Government, Korea.